RAISE-26

Mirror, Mirror on the Wall, Is AI Transforming Us All?

The **Attention** Heads

31 January 2026

Motivation & Hypothesis

The Question: How does AI transform human behavior?

The Answer: Not through direct technology use, but through the media mirror that shapes how we perceive AI.

Our research examines 10,000+ news headlines across four media tiers (June-Sept 2025) to reveal a disturbing pattern: *media systematically amplifies AI optimism while suppressing critical discourse about risks, displacement, and societal impact.*

The Transformation: We're not being changed by AI itself; we're being conditioned by a broken mirror that shows us only what powerful interests want us to see.

Broken Mirror Hypothesis

The Assumption: Media is a passive reflection of technological progress.

Our Discovery: The mirror is *broken*; it amplifies AI benefits while systematically containing/suppressing human risks.

We are being transformed by media-manufactured optimism.

Research Roadmap & Methodological Pivot

1

The Scientific Pivot

90-day window is a not a longitudinal study – **direct behavioral transformation** is not provable.

We analyze the **Cause** (The Information Engine) rather than the **Effect** (Behavioral Change).

Identifying how the "**Mirror**" is engineered before it reaches the public.

2

Analytical Roadmap

Sentiment **Cascades**

Why **Optimism** Cascades

Narrative **Normalization**

Semantic Audit

Restoring the Mirror

3

What We Test

Does elite news **predict** public sentiment?

Measuring "**tell-ability**" of negative vs. positive news.

Tracking the **repetition and stability** of dominant themes.

Mass-scale validation of 10,500 headlines via LLM.

Can we prove AI transformation by analyzing the **Narrative Engine** that manufactures it?

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Multi-Method **Causal Evidence** Pipeline for **Sentiment Cascades**

Dataset:

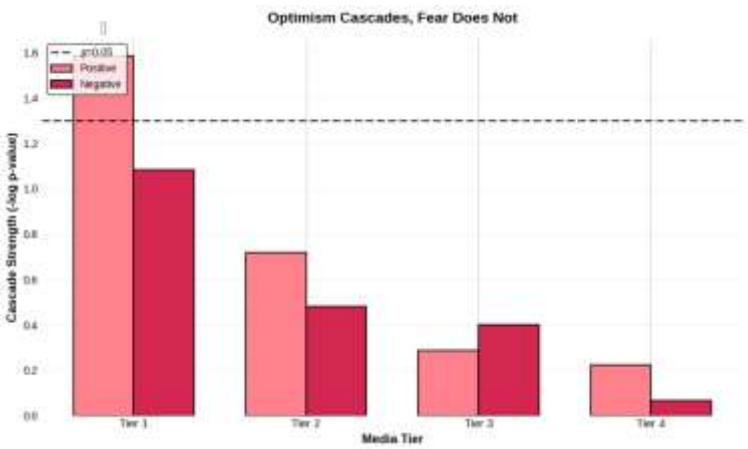
- 10,500+ headlines (June-Sept 2025)
- **Tier 1** (Elite Tech): TechCrunch, Wired, The Verge, Ars Technica
- **Tier 2** (Mainstream News): NYT, WSJ, BBC, CNN, Reuters, Bloomberg, Forbes
- **Tier 3** (Trade/Niche): Yahoo Finance, Business Insider, ZDNet, MarketWatch
- **Tier 4** (Other/Aggregators): All remaining sources
- Academic sources filtered

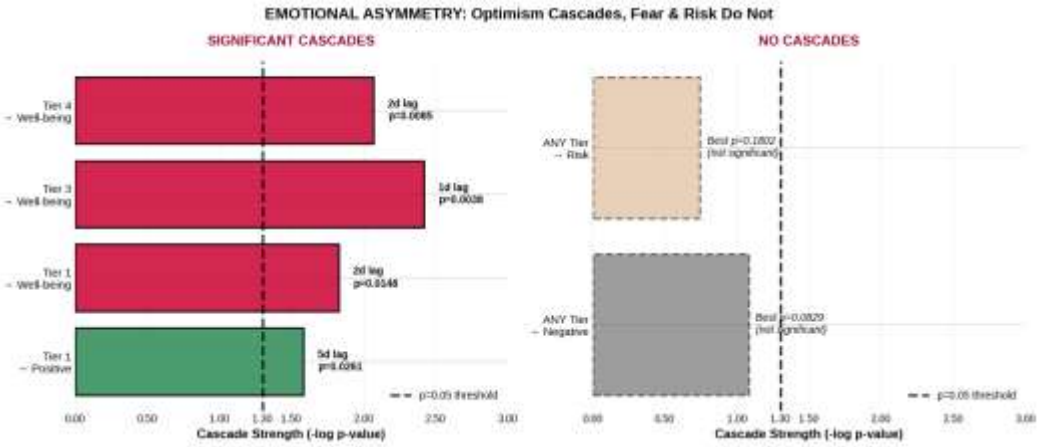
Methods:

- **VAR Models:** Lag optimization (AIC/BIC), tested 1-10 days
- **Granger Causality:** All 16 tier pairs, lags 1d/2d/5d/7d, $p<0.05$
- **CCM:** Library sizes 10-120, $E=3$, $\tau=1$
- **Dimensions:** Sentiment, well-being, risk, creativity, work, human roles, social interaction

The **Causal Evidence** demonstrate that optimism cascades significantly while fear is systematically contained; media manufactures consensus rather than reflects reality.

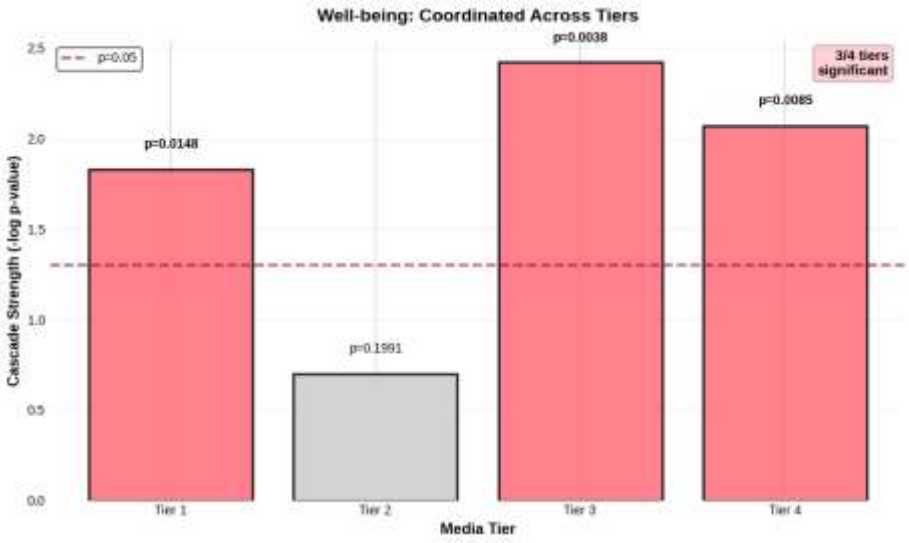
Emotional Asymmetry: Statistical Proof of Coordinated Networks

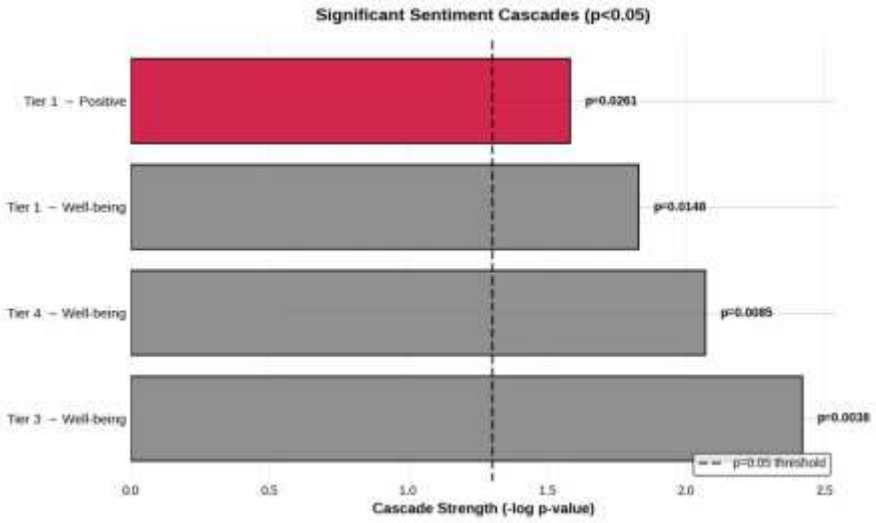




Key Insight: Positive cascades show declining strength across tiers (traditional cascade model), but negative sentiment has NO cascade from *any* tier — this contradicts passive information flow.

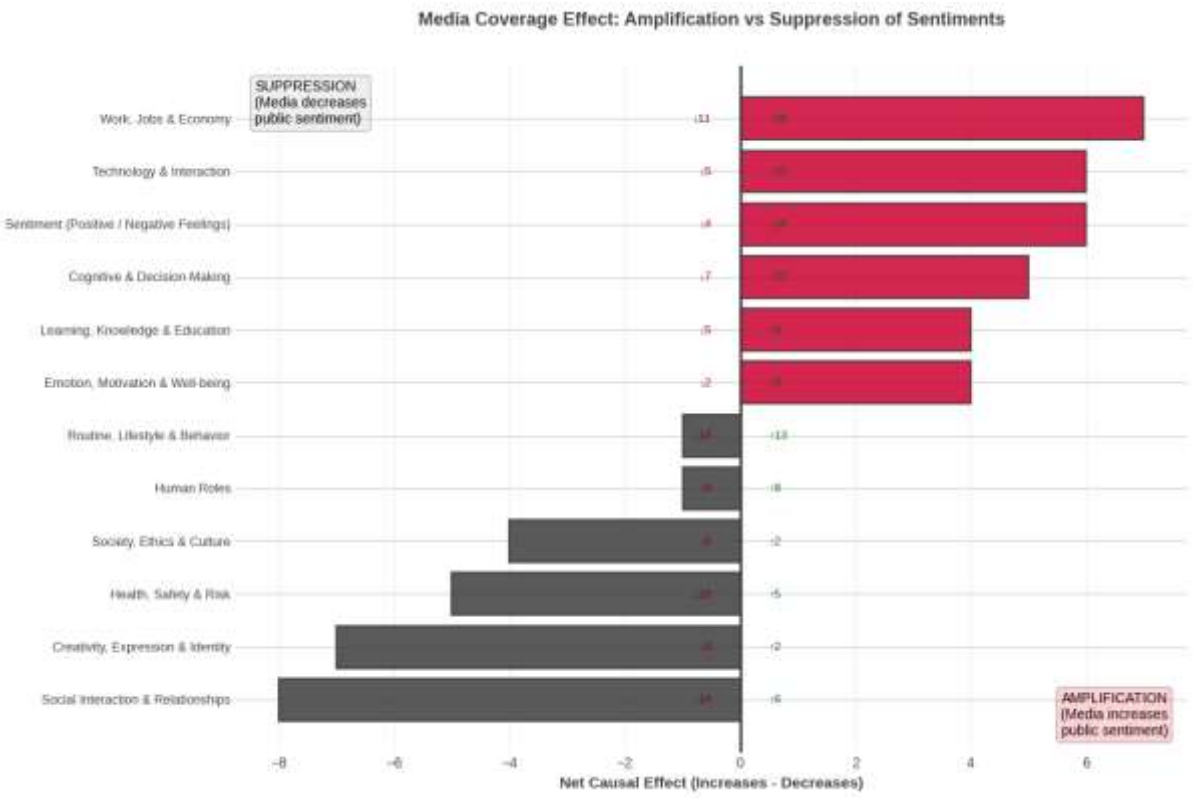
Power Hierarchy: Business Media > Tech Media for **Sentiment Cascades**





Key Insight: Mainstream business media (Tiers 3/4) drive public sentiment **3×** stronger than tech sources—the Mirror reflects economic interests, not technological expertise per se.

Systematic Filtering: Media's Asymmetric Coverage Effects

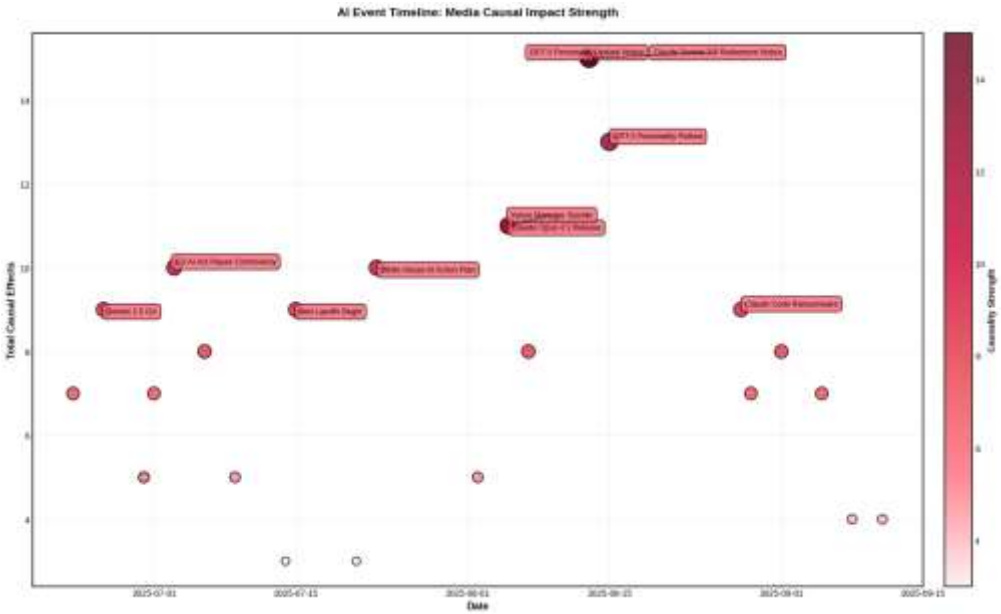


Key Insight:

χ^2=15.82, p<0.01

Work/economy amplified +18, social interaction suppressed -14, less than 0.01% chance this pattern is random,

Event-Driven Analysis: Tracking AI Announcement Impacts



What the Timeline Shows: Bubble color intensity = causality strength. Major coordination events (Gemini 2.5, GPT-5, Yahoo! AI) show strongest cross-tier influence, revealing synchronized sentiment engineering patterns.

25 Major AI Events (June–September 2025)

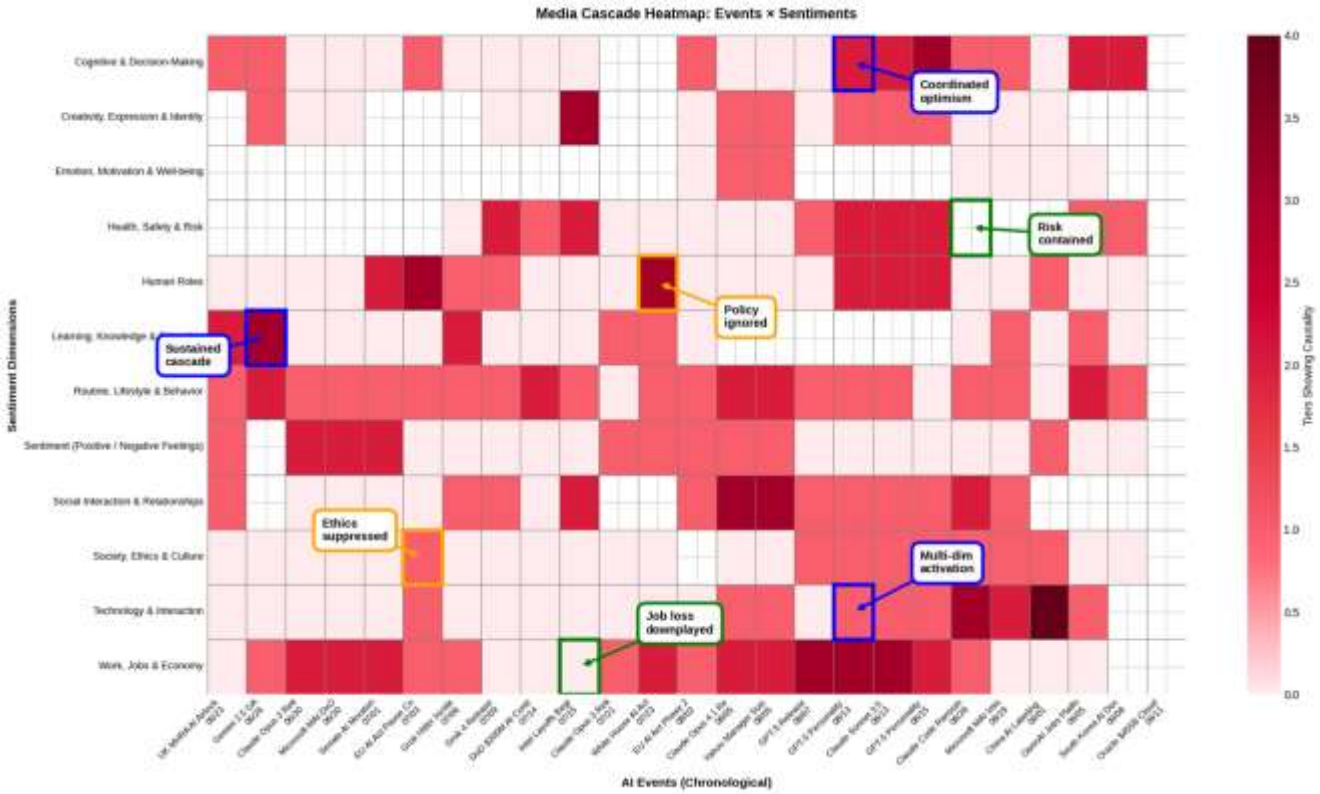
Selection Criteria:

- Product launches, policy announcements, breakthrough research, major incidents
- Events with significant media attention across multiple tiers
- 14-day analysis window around each event (±7 days)

Key Events Analyzed:

- Product: Claude Sonnet 3.5, Gemini 2.5, GPT-5 Personality Rollout, Yahoo! AI Shopping
- Policy: White House AI Action Plan, EU AI Act Pause, Intel layoffs
- Security: Claude Code ransomware attack
- Research: Breakthrough discoveries and retirement notices

How AI Announcements Trigger **Sentiment Cascades**



Blue boxes (Coordinated optimism): Product launches create dark vertical bands—multi-dimensional activation cascading across tiers.

Green boxes (Isolated spikes): Security/job events produce moderate red cells—concern acknowledged but downplayed, not amplified.

Orange boxes (Regulatory silence): Policy events show pale cells where strong ethical coverage should exist—critical discourse systematically suppressed.

Takeaways from **Causal Evidence**

- **Zero negative cascades** from any tier – fear doesn't flow, optimism does (*coordinated amplification*)
- Business media drives sentiment **3× stronger** than tech sources, i.e. the mirror reflects economic interests, not technical reality
- Work/economy amplified **+18** effects; social concerns suppressed -14 (χ^2=15.82, $p<0.01$) – *systematic filtering*
- Product launches = durable cascades; risk events = ephemeral containment

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2 Analytical Roadmap

- Sentiment **Cascades**
- Why **Optimism** Cascades
- Narrative **Normalization**
- **Semantic Audit**
- **Restoring** the Mirror

3 What We Test

- Does elite news **predict** public sentiment?
- Measuring "**tell-ability**" of negative vs. positive news.
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Can we prove AI transformation by analyzing the **Narrative Engine** that manufactures it?

Hypothesis Testing: Why does Optimism Cascade?

In general, there could be multiple reasons as to why positive news percolates down the chain faster than bad news about AI. We systematically test and present evaluation of 4 hypotheses:

H1: Attention Economy: Positive headlines use more “clickbait” language

H2: Advertiser Pressure: Ad-dependent sources show more positive coverage

H3: Ownership Conflicts: Tech billionaire-owned media shows bias

H4: Narrative Complexity: Negative stories are harder to sell and share

H1: **Attention Economy**

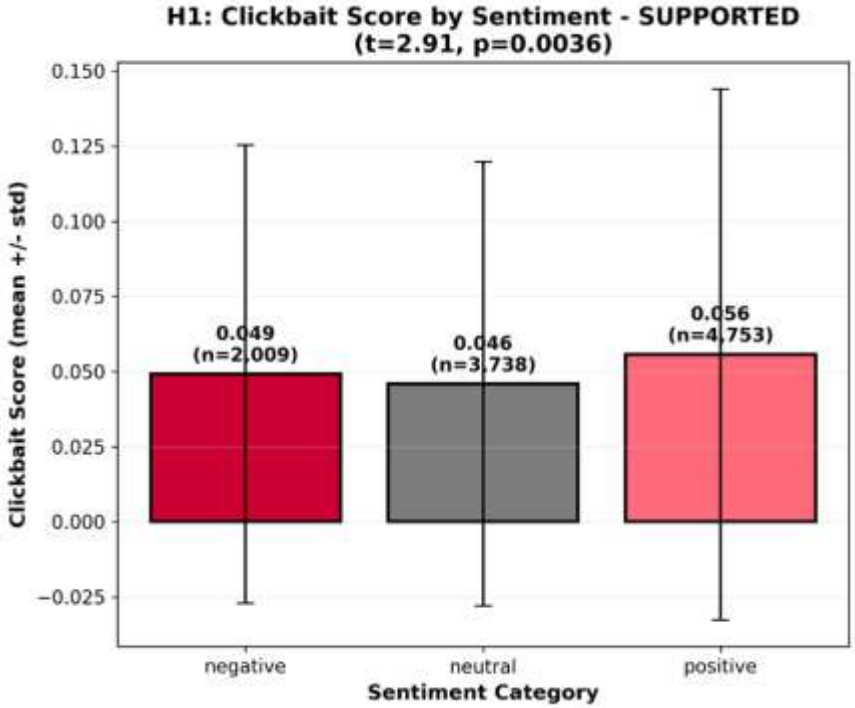
H1: Hypothesis: If the attention economy drives asymmetric propagation, positive headlines should have HIGHER clickbait scores (more engagement-optimization).

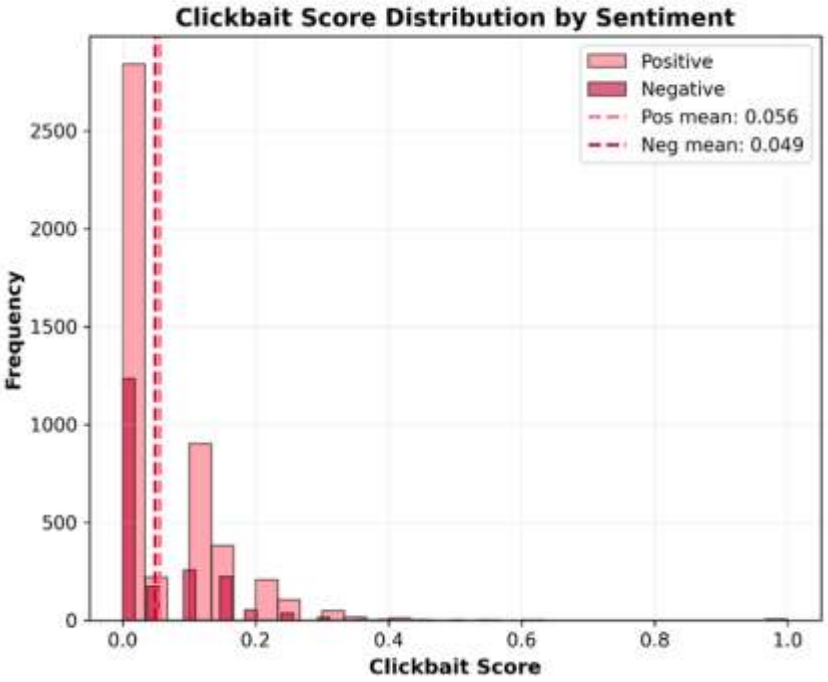
1. Compute *VADER* sentiment analysis on headlines
2. Compute clickbait scores based on linguistic features
3. Compare scores between positive, negative, and neutral sentiment headlines
4. Use independent samples t-test for statistical significance

Clickbait Scoring (based on Potthast et al. 2016, Chakraborty et al. 2016)**:**

- Exclamation marks: +0.20 each (creates urgency)
- Question marks: +0.15 each (curiosity gap)
- Numbers in title: +0.10 (listicle format)
- Superlatives: +0.15 each (emotional amplification)
- Clickbait phrases: +0.20 each (curiosity exploitation)
- ALL CAPS words: +0.10 each (visual emphasis)
- Quotes: +0.05 (dramatic effect)

H1: **Attention Economy**



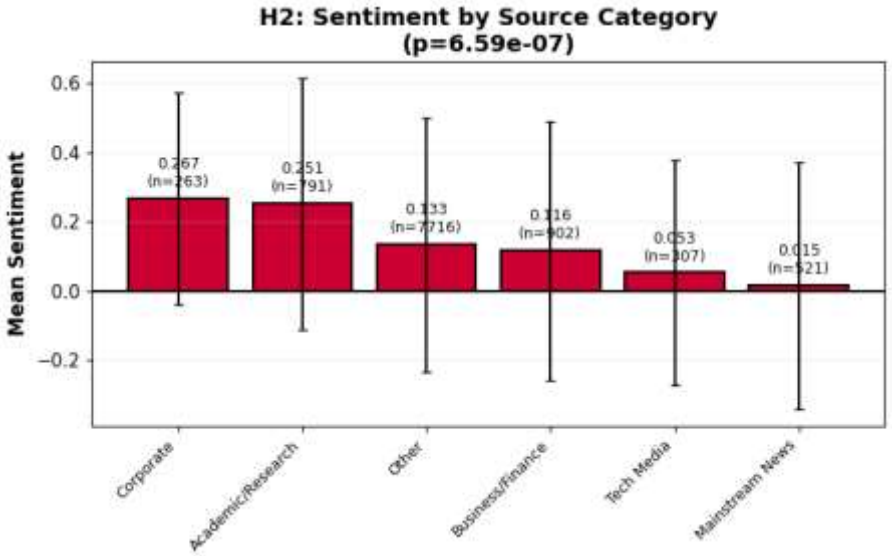


Key Insight: Clickbait is statistically significant but practically negligible, i.e., the effect is real but far too small to explain the propagation gap.

H2: **Advertiser Pressure**

H2: Hypothesis: If advertiser pressure drives bias, sources with high tech ad revenue (business media, tech media) should show more positive sentiment than mainstream news outlets with diverse revenue streams.

1. Compute *VADER* sentiment analysis on headlines
2. Categorize sources by type (Corporate, Academic, Tech Media, Business/Finance, Mainstream News, Other)
3. Compare mean sentiment (*VADER* compound) across categories
4. Test whether ad-dependent outlets show more positive coverage

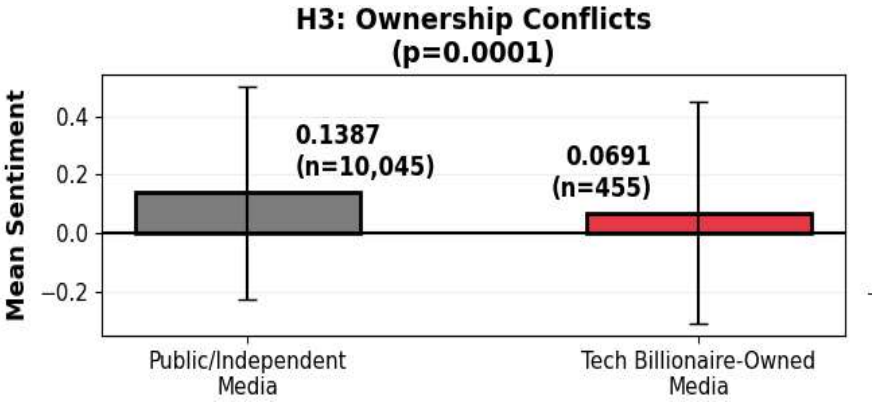


Key Insight: The asymmetry comes from supply, not editorial pressure, positive PR content is free, critical investigation is expensive.

H3: **Ownership Conflicts**

H3: Hypothesis: If ownership creates editorial bias, outlets owned by tech billionaires should show *more* positive sentiment than independent media.

1. Compute *VADER* sentiment analysis on headlines
2. Identify outlets owned by tech billionaires
3. Compare sentiment between owned vs independent/public outlets
4. Use independent samples t-test for statistical significance



Key Insight: Tech billionaire-owned outlets are actually *more* critical of AI. Ownership does ***not*** seem to create bias in this specific dataset.

H4: **Narrative Complexity**

H4: Hypothesis: If narrative complexity exists, negative headlines should have *more* abstract language, hedging phrases, and complex structures than positive headlines, explaining propagation differences without invoking editorial bias.

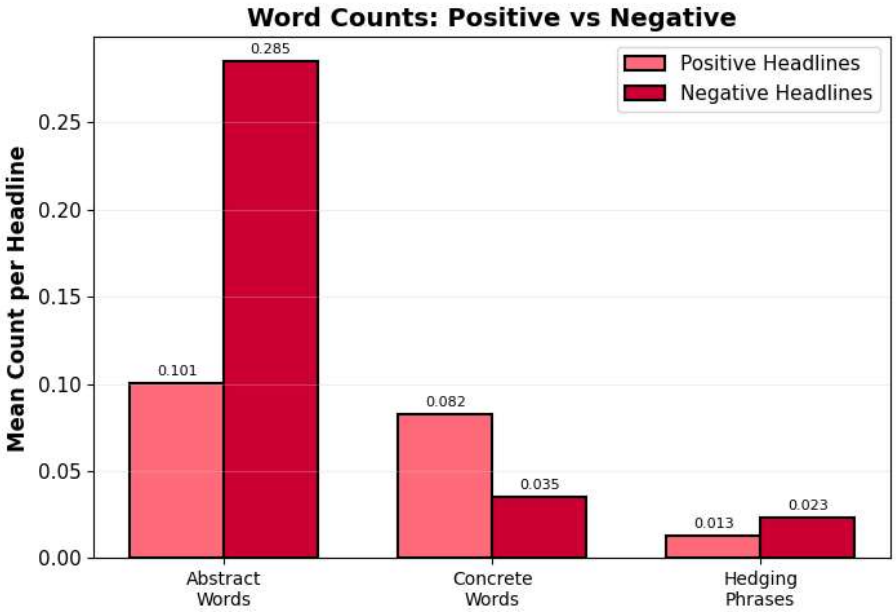
Theoretical Foundation:

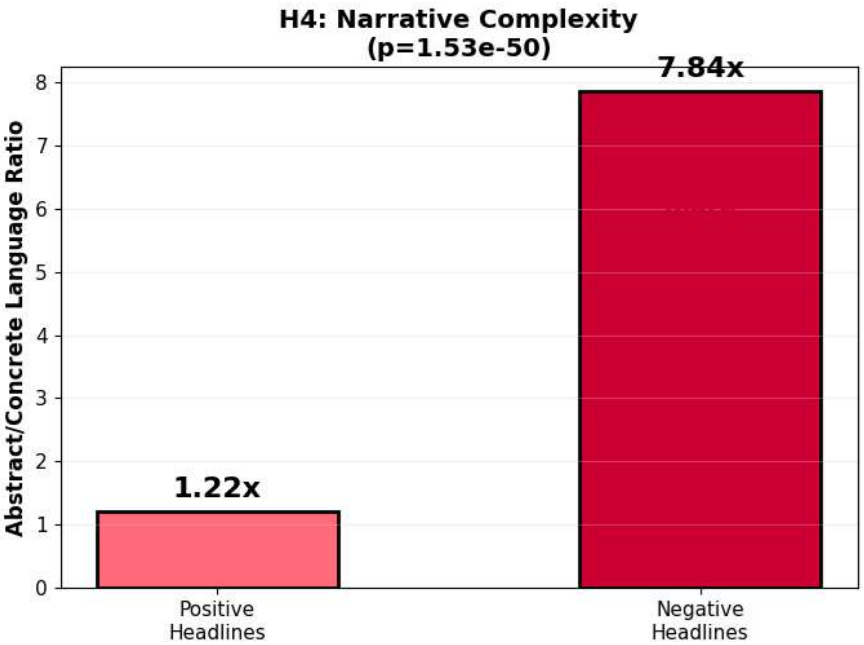
- Dual Coding Theory (Paivio, 1986): Concrete words create mental images, making them more memorable and shareable
- Complex content doesn't fit tweet/headline formats

Key Metrics:

1. Abstract Language: could, might, potentially, concerns, implications, etc.
2. Concrete Language: launches, announces, acquires, builds, delivers, etc.
3. Abstract/Concrete Ratio: Key measure of structural complexity
4. Hedging Phrases: "if", "whether", "could be", "raises questions", etc.
5. Complexity Score: Weighted combination of all factors

H4: **Narrative Complexity**





Key Insight: Negative headlines use 3x more abstract words and half the concrete words. There is a 6.4x difference in how negative headlines present as abstract or vague.

Takeaways from **Hypothesis Testing**

- **H1 (Attention Economy):** Statistically significant but tiny effect—clickbait doesn't explain the gap
- **H2 (Advertiser Pressure):** Corporate PR is most positive; bias comes from content supply, not editorial pressure
- **H3 (Ownership Conflicts):** Tech billionaire-owned outlets are more critical—ownership does not seem to cause this particular bias.
- **H4 (Narrative Complexity):** A strong support for negative headlines being more *abstract*, likely to appear as a ‘may be’ or ‘could be’ with *fewer* concrete negative headlines.

Bottom line: The mirror is broken by *structure*, not conspiracy. Simple stories propagate; complex stories stay contained.

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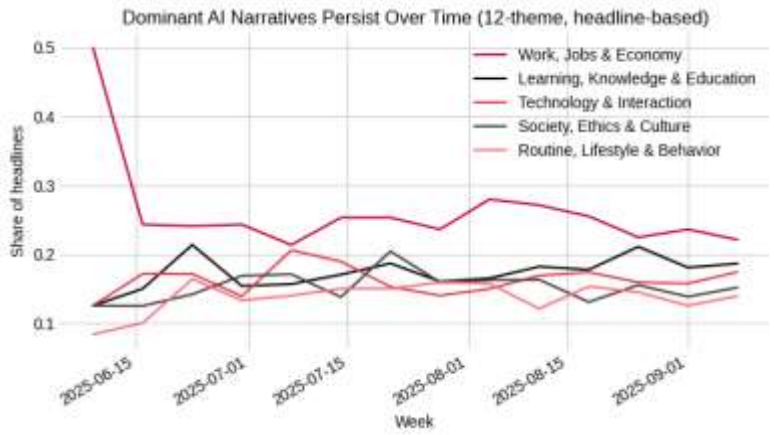
Narrative **Normalization**: Repetition as Reality

We operationalize **Narrative Normalization** through the following mechanisms:

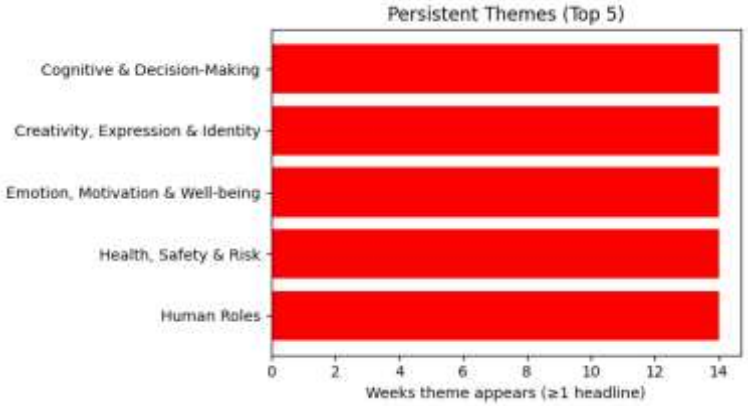
- **M1: Dominance**
  A small set of narratives consistently occupies a large share of media coverage.
- **M2: Persistence**
  The same narratives recur week after week, independent of changing AI events.
- **M3: Attention Concentration**
  Media attention disproportionately focuses on a few dominant narratives.
- **M4: Attention Inequality**
  Narrative diversity remains bounded over time, indicating structural imbalance rather than rotation.

Narrative Normalization: **Repetition as Reality**



Dominant Narrative: Narratives that consistently occupy a large share of media coverage, measured by the proportion of weekly headlines

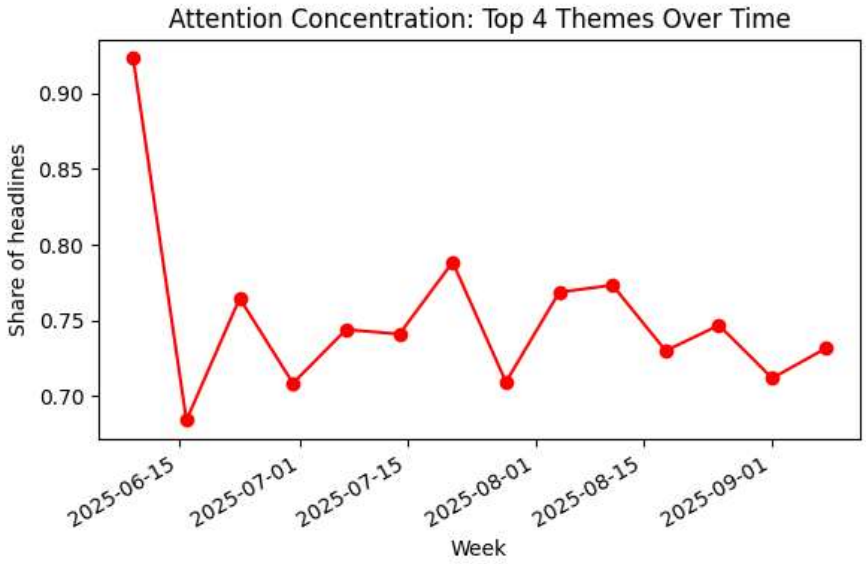
Persistent Narrative: Narratives that recur across many weeks, regardless of short-term fluctuations in coverage volume



Key Insight: Media coverage repeatedly concentrates on a small set of narrative that are both **dominant** and **persistent**, despite shift in AI events

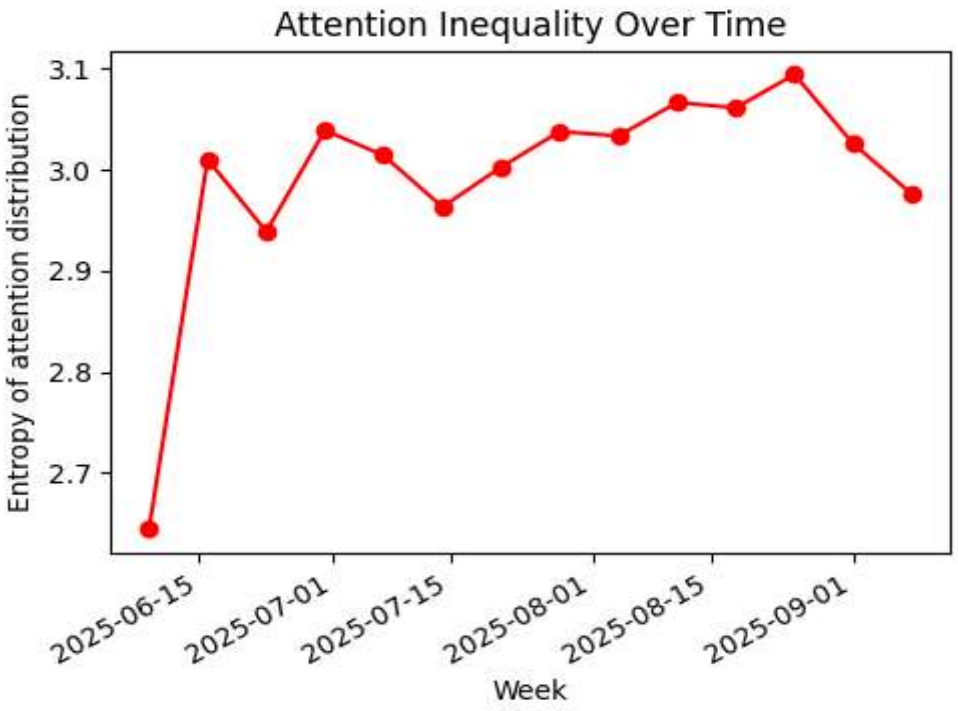
Narrative Normalization: **Attention Concentration Over Time**

Dominant Themes (Top 4):
1) Work, Jobs, & Economy
2) Learning, Knowledge & Education
3) Technology & Interaction
4) Society, Ethics & Culture



Key Insight: The top themes capture most of the attention. Dominance is shown to vary but still does not go less than 50%.

Narrative Normalization: **Attention Inequality**



Dominant Themes (Top 4): 1) Work, Jobs, & Economy
2) Learning, Knowledge & Education
3) Technology & Interaction
4) Society, Ethics & Culture

Key Insight: Entropy quantifies attention diversity; its stability shows that AI coverage remains concentrated rather than diversifying over the weeks.

Takeaways from Narrative **Normalization**

- **Repetition over novelty:** A small set of AI narratives persist across weeks, independent of changing events.
- **Structural dominance:** These narratives consistently occupy a disproportionate share of media coverage.
- **Concentrated attention:** The majority of AI-related attention is captured by just a few dominant themes.
- **Stable inequality:** Attention diversity does not increase over time; entropy remains bounded and stable.
- **Normalization effect:** Repeated exposure to the same frames establishes a “background reality” through which AI is interpreted

Bottom line: Media does not mirror reality, it repeatedly amplifies a narrow set of narratives, shaping how AI is understood.

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Verification: The LLM Semantic Audit

Analysis Pipeline

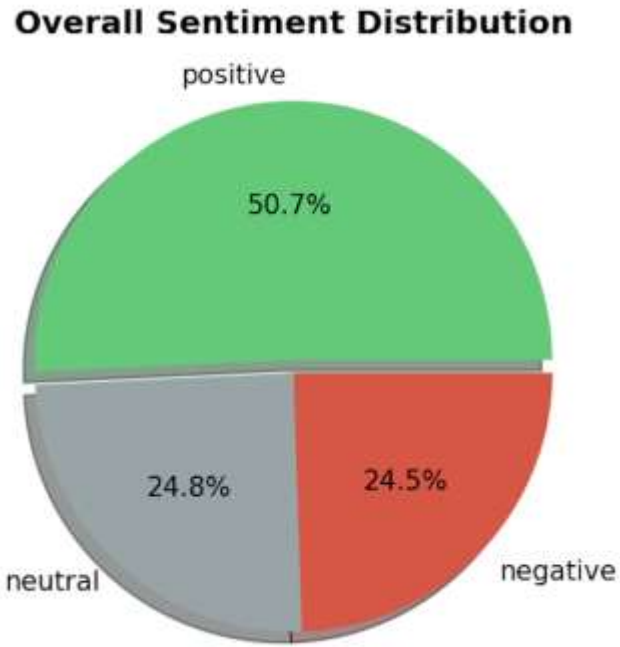
1. **Headline Input**
   10,500 news title
2. **Structured Prompt**
   Predefined taxonomy
3. **LLM Classification**
   Llama-3.1-8B-Instruct
4. **JSON Output**
   Llama-3.1-8B-Instruct
5. **Aggregation**
   Statistical analysis
6. **Deep Interpretation**
   LLM analysis statistics

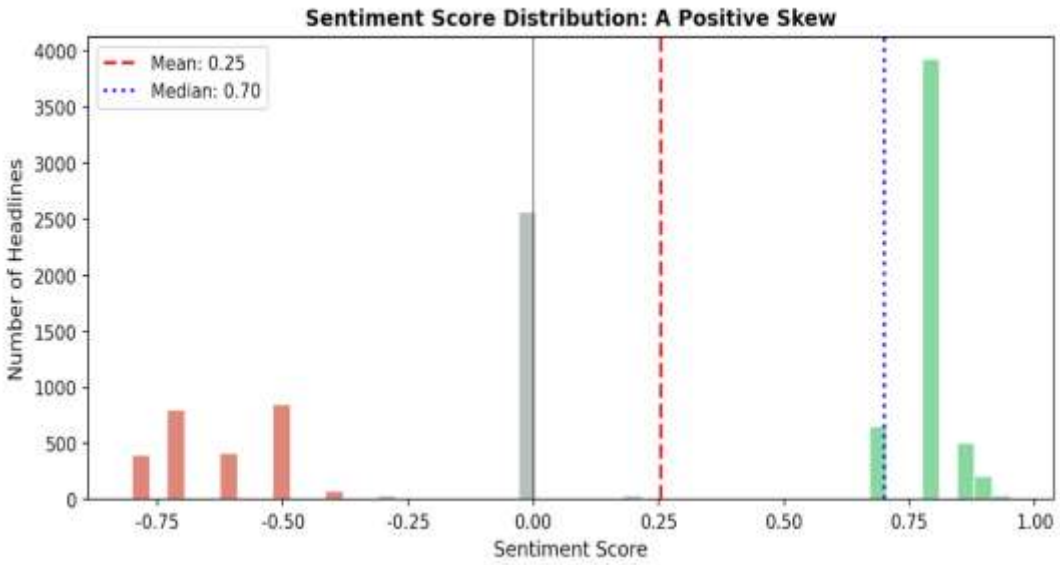
Compressed Prompt Example

```
You are an expert media analyst specializing in AI news
coverage. Analyze the following AI-related news headline
comprehensively.

HEADLINE: {headline}

Output a JSON object with:

- sentiment ∈ {positive, neutral, negative} and
sentiment_score ∈ [-1, 1]
- framing ∈ {opportunity, threat, neutral_informative,
hype, critical, solution, disruption}
- tone ∈ {sensational, measured, technical, casual,
alarming, optimistic}
- key_themes: 1-3 items from a predefined theme list
{theme}
- emotional_words: up to 5 emotionally charged tokens
- rhetorical_devices ∈ {question, quote, superlative,
numbers, metaphor, urgency, negation, personification}
- bias_indicators ∈ {corporate_promotion,
fear_mongering, tech_utopia, balanced, expert_appeal,
anecdotal}
- confidence ∈ [0, 1]
```

```
You are a critical media studies scholar analyzing
aggregate patterns in AI news coverage.

INPUT:
- Aggregate statistics from LLM-based semantic
classification of AI-related news headlines
{statistics}

TASK:
Interpret the statistical findings to explain systematic
patterns in AI media coverage.

Produce a structured analysis with:
- 3-4 key insights explaining sentiment skew, framing
imbalances, and outlet-level differences
- Root cause analysis of media bias:
  {corporate PR influence, journalist cognitive bias,
traffic-driven incentives}
- 2-3 consequences for public understanding:
  {AI risk perception, democratic deliberation, informed
consent}
- Assessment of evidential strength for the "Broken
Mirror" thesis:
  {supporting patterns, alternative explanations,
missing evidence}
- 3-4 actionable policy recommendations for:
  {media organizations, policymakers, researchers,
public}

Write in an academic but accessible tone and explicitly
reference the provided statistics.
```

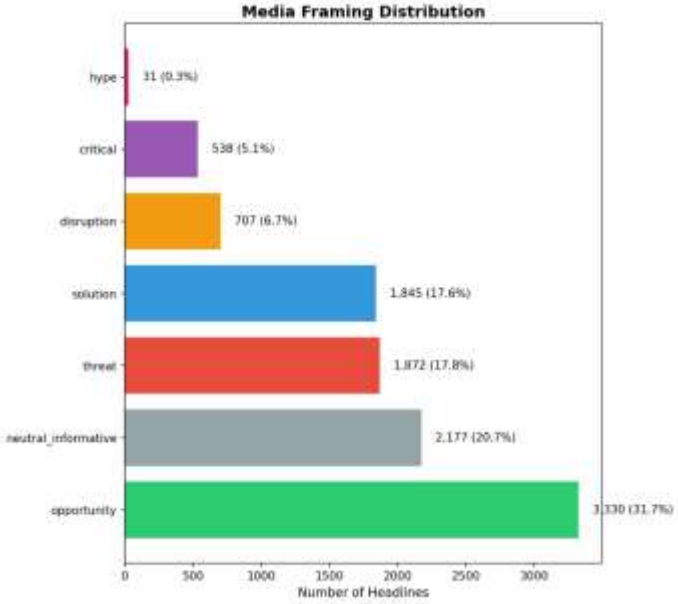
Headline Sentiment Analysis

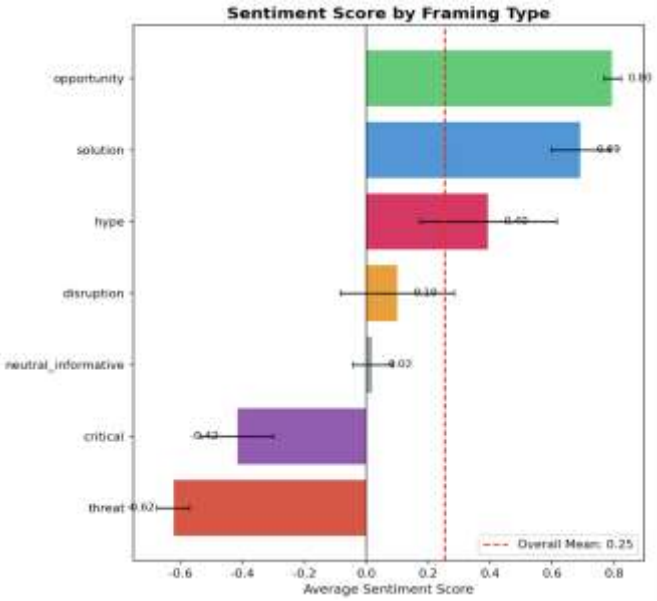




LLM Analysis: Analyzing 10,500 headlines, we uncover a systematic positivity bias: **50.7% positive vs. 24.5% negative** (2.05:1). Sentiment scores concentrate around **+0.7**, showing that the media mirror doesn't just reflect reality—it **magnifies optimism**. This directly reinforces our earlier finding on **optimism cascades** (Slides 6/14).

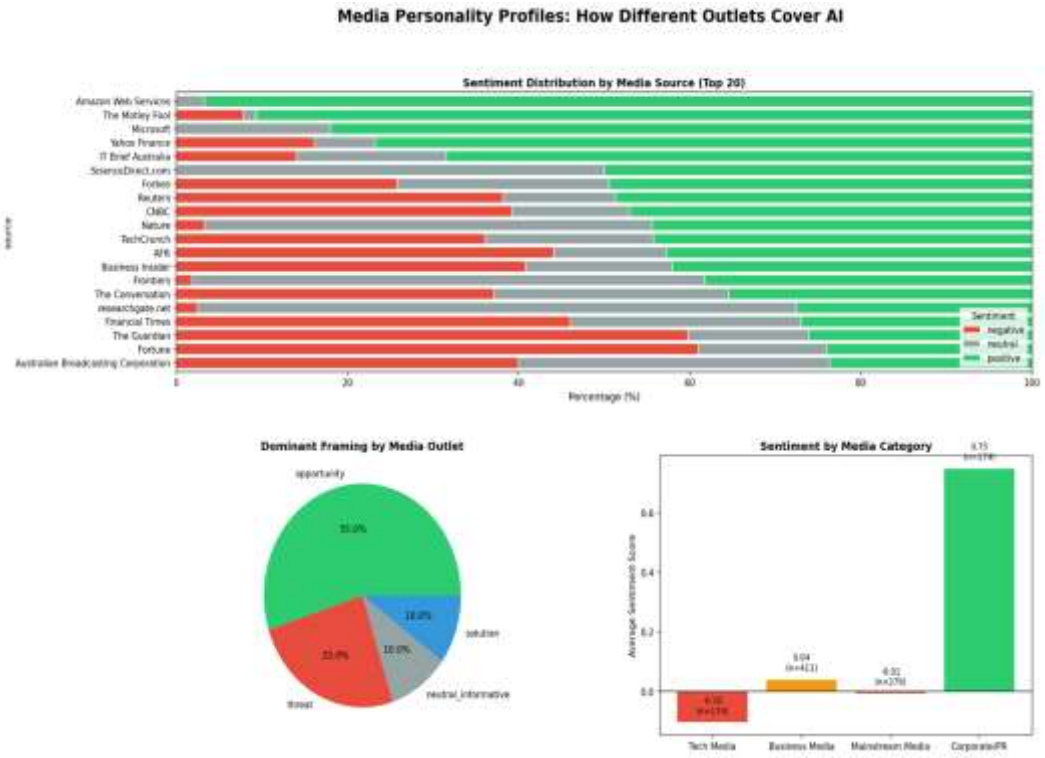
Media Framing Pattern Analysis





LLM Analysis: With **31.7%** of headlines framing AI as an opportunity and only **17.8%** as a threat, media narratives already favor optimism **1.8:1**. When *solution* framing is added, positive narratives overwhelm risk narratives **2.8:1**. This is not a neutral imbalance: it reflects a media logic that **systematically selects the hopeful story**, while marginalizing discussions of risk.

Media Source Analysis

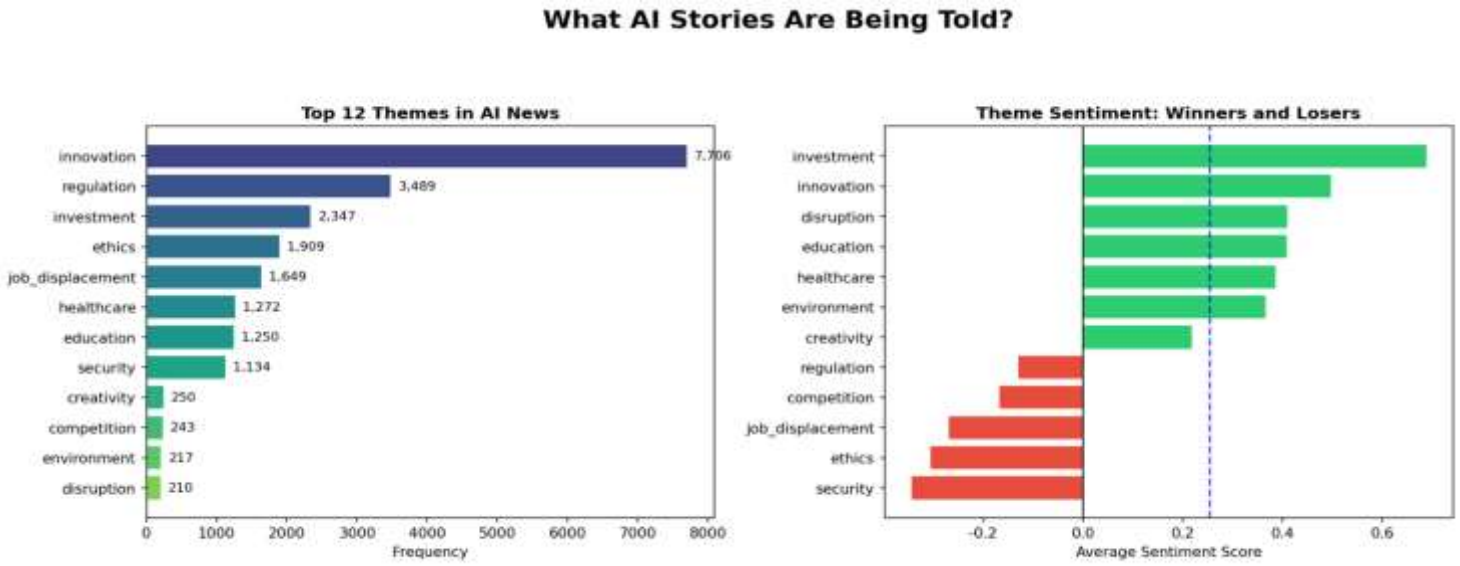


LLM Analysis: The sentiment gap between **corporate/PR sources (+0.75)** and **tech media (−0.10)** reveals a pronounced optimism bias in corporate-aligned coverage.

This disparity suggests that **corporate interests may be shaping media narratives**, potentially influencing public opinion toward AI adoption.

This finding **aligns with our hypothesis test on advertiser pressure** (Slide 17).

Theme Analysis



LLM Analysis: *Innovation* dominates AI coverage (**7,706 mentions**), while *job displacement* (**1,649**) and *ethics* (**1,909**) remain peripheral. Sentiment follows the same asymmetry: *investment* stories are highly positive (**+0.65**), whereas *security* (**−0.15**) and *ethics* (**−0.10**) skew negative.

The result is a narrative economy where **good news travels farther than warnings**, systematically sidelining AI risks.

This LLM evidence is **consistent with the asymmetric media coverage effects observed in our causal analysis** (Slide 9).

Research Roadmap & Methodological Pivot

1

The Scientific Pivot

90-day window is a not a longitudinal study – **direct behavioral transformation** is not provable.

We analyze the **Cause** (The Information Engine) rather than the **Effect** (Behavioral Change).

Identifying how the "**Mirror**" is engineered before it reaches the public.

2

Analytical Roadmap

Sentiment **Cascades**

Why **Optimism** Cascades

Narrative **Normalization**

Semantic Audit

Restoring the Mirror

3

What We Test

Does elite news **predict** public sentiment?

Measuring "**tell-ability**" of negative vs. positive news.

Tracking the **repetition and stability** of dominant themes.

Mass-scale validation of 10,500 headlines via LLM.

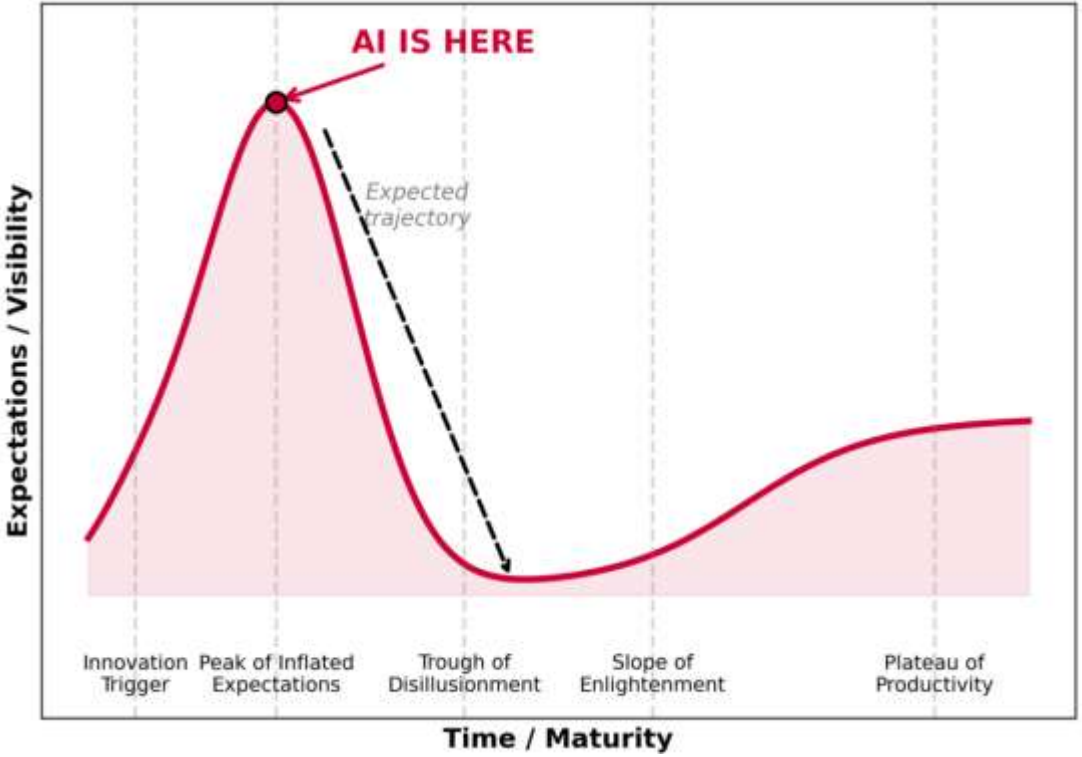
Can we prove AI transformation by analyzing the **Narrative Engine** that manufactures it?

What does this mean for AI and the *broken* mirror?

Our sentiment analysis shows overwhelmingly positive coverage for AI.

This asymmetry places AI squarely at the *peak of inflated expectations* in the gartner hype cycle.

If history is any guide, a correction is coming. The trough of disillusionment typically follows when implementations *fail* to meet *expectations.*



Policy Synthesis: The "So What?"

<u>The Problem:</u> The mirror isn't broken by malice; it's broken by **structure**. Media structures **amplify, suppress and normalize selective narratives**, shaping our perception of AI.

The Strategic Fixes:

- **Transparency:** Standardized disclosure of AI-related content originating from corporate or institutional sources to support informed interpretation
- **Structural Support:** Sustained funding for in-depth and investigative AI journalism to ensure complex and risk-oriented narratives remain viable.
- **Algorithmic Oversight:** Regular sentiment-balance and diversity assessments for major news aggregation and recommendation systems.
- **System Monitoring:** Analysis of media dynamics to better understand narrative propagation patterns

Effective AI governance requires attention not only to technology, but to the media systems through which AI is understood.

Key Takeaways

- AI is not the primary driver of change; media framing is.
- Positive AI narratives propagate; risk narratives do not.
- Attention concentrates on a small set of themes.
- Narrative diversity does not increase over time.
- Narrative complexity limits the visibility of risk.
- AI governance must account for media systems.

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